

Data_Analyst — Step 1

Learn Excel & SQL

Detailed Learning Notes • CareerCompass

1. Learning Objective

Become effective at spreadsheet analysis and database querying for common business questions.

2. Core Concepts — Detailed Breakdown

Excel foundations

Learn tables, sorting/filtering, formulas, relative/absolute references, XLOOKUP, conditional formatting, pivot tables and charts. Use consistent data types and avoid manual copy-paste workflows.

SQL querying

Master SELECT, WHERE, ORDER BY, GROUP BY, HAVING, joins, CASE, subqueries and CTEs. Understand row grain so aggregates are correct.

Business metrics

Translate questions into measurable definitions such as revenue, conversion rate, retention or average order value. Always specify numerator, denominator and time period.

Data quality

Check duplicates, nulls, invalid categories and unexpected joins before trusting a result.

3. Recommended Learning Workflow

- Practice Excel on a clean dataset.
- Write basic SQL queries.
- Add joins and grouped metrics.
- Recreate the same analysis in SQL and Excel.
- Document metric definitions.

5. Hands-on Project / Practice

Analyze an e-commerce dataset. Build an Excel workbook with cleaning, pivot analysis and charts, then reproduce key metrics with SQL. Reconcile the outputs and investigate any mismatch.

6. Common Mistakes & Pitfalls

- Using merged cells as a data structure.
- Hard-coded totals.
- Wrong join keys.
- Using COUNT when COUNT DISTINCT is needed.
- Not defining the metric grain.

7. Completion Checklist

- I can build a clean Excel table.
- I can use XLOOKUP and pivot tables.
- I can write multi-table SQL.
- I can define business metrics precisely.
- I can reconcile results across tools.

8. Interview / Assessment Questions

- When would you use a pivot table?
- What is the difference between COUNT and COUNT DISTINCT?
- Why can a join inflate revenue?
- What is an absolute reference in Excel?
- How do you define a conversion rate?

9. Recommended References

Data Analyst Bootcamp: <https://www.youtube.com/watch?v=PSNXoAs2FtQ>

Microsoft data analytics: <https://learn.microsoft.com/en-us/training/paths/data-analytics-microsoft/>

Microsoft Excel Training: <https://support.microsoft.com/excel>

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.